

Missing Data and Imputation Analysis

The dataset contained 24 subjects and 152 craniofacial measurement variables. Overall, 238 of 3,648 measurement cells were missing (6.52%). Missingness was concentrated rather than uniformly distributed: 51 measurements contained at least one missing value, whereas 101 were complete. Twelve subjects had missing measurements and the missing-data heatmap showed clear blocks of co-missing values. Several measurements also shared similar missingness patterns across the same subjects, indicating that missingness sometimes occurred in groups rather than as isolated values. The missingness mechanism was investigated using the available subject characteristics. There was no significant relationship between age and the number of missing measurements (Spearman $\rho=0.27$, $p=0.20$), no significant difference in age between subjects with and without missing data (Wilcoxon $p=0.40$), and no significant association between gender and missingness status (Fisher's exact test, $p=0.22$). These findings do not confirm MCAR. The structured missingness pattern suggests that MCAR should not be assumed, while MAR cannot be excluded because missingness may depend on other observed measurements not assessed here. MNAR also cannot be ruled out because the reasons for missingness and the unobserved values are unknown. The block-missingness pattern may reduce the information available for imputation when several measurements are missing simultaneously, so these imputations should be interpreted more cautiously.

Median imputation was implemented as a simple baseline, while k-nearest-neighbour (kNN) imputation was selected as the main candidate method. kNN was considered suitable because many craniofacial measurements were correlated, allowing information from similar subjects to contribute to imputation. Measurements were standardised for distance calculation so that variables with larger numerical scales did not dominate neighbour selection. For each missing value, distances were calculated using measurements observed in both the recipient and potential donor and the missing value was replaced by the mean of up to k nearest donors with an observed value for the target measurement. Candidate values of $k = 3, 4$ and 5 were compared using the same validation dataset; $k = 3$ was selected because it produced the lowest RMSE, while MAE was also examined as a complementary error measure. No subjects or measurement variables were excluded from imputation because none were completely missing and all originally observed values were preserved.

Validation was performed by temporarily masking approximately 10% of observed values in variables that originally contained missingness. This produced 92 artificial missing values with known true values, while the original missing observations remained missing. Median imputation produced an RMSE of 6.68 and an MAE of 4.94, compared with an RMSE of 3.82 and an MAE of 2.68 for kNN with $k = 3$. kNN also produced the smaller absolute error for 76 of the 92 masked observations, compared with 16 for median imputation. Performance was also examined across variables and different levels of original missingness; kNN generally performed better, although estimates for some variables were based on very few masked observations and should therefore be interpreted cautiously. In addition, kNN better preserved the original correlation structure: across 4,054 originally strong correlation pairs, the mean absolute change in correlation was 0.016 for kNN compared with 0.091 for median imputation. Distributional checks also indicated that median imputation tended to reduce variability, whereas kNN generally preserved the observed means and standard deviations more closely.

Based on the validation results, kNN with $k = 3$ is recommended for exploratory analysis of datasets similar to the present sample because it produced lower RMSE and MAE than median imputation and better preserved the correlation structure and variability of the craniofacial measurements. Imputation should only be performed when the target variable has sufficient observed donor values and the subject shares enough observed measurements with potential donors for a meaningful distance calculation. Variables with very high missingness, subjects with extensive or block-pattern missingness, and cases with fewer than three suitable donors should be treated as warning conditions requiring manual review rather than routine automatic imputation. Each imputed value should be clearly flagged and accompanied by information on the imputation method, selected k , number of usable donors and, where feasible, neighbour distances to indicate the level of support for the estimate. kNN may produce unreliable or misleading results when few suitable neighbours are

available, many measurements are missing simultaneously, or future subjects differ substantially from the reference sample. Therefore, although kNN with $k=3$ is suitable for exploratory and internal research use, it should not yet be used as an unrestricted clinical or production method; clinical deployment would require validation on a larger independent dataset, prospectively defined data-quality thresholds and manual review or non-imputation of low-support cases.